

- 25** Sickle cell anaemia is caused by a mutation in an allele of the gene that codes for the β -globin polypeptide of haemoglobin.

The diagram shows the sequence of bases in a small section of the template strand of DNA for both the Hb^A (normal) and Hb^S (sickle cell) β -globin alleles.

Hb^A CTGACTCCTGAGGAGAAGTCT

Hb^S CTGACTCCTGTGGAGAAGTCT

Both the polypeptides for Hb^A and Hb^S are the same length.

How will the mutation in the allele result in the production of an altered version of the β -globin polypeptide?

- A** A tRNA molecule with the anticodon GUG will form hydrogen bonds with the altered codon on mRNA.
- B** All the amino acids coded for after the mutation will differ from those in the Hb^A protein.
- C** mRNA transcribed from the Hb^S allele will contain the codon CAC instead of the codon CTC.
- D** The ribosome will be unable to continue translation of the Hb^S mRNA after the altered codon.